WORKED SOLUTION

STEP II 1994, Question 14

Probability and Statistics • Geometrical Probability

STEP II 1994, Question 14

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/94-s2-q14

The key idea

"Uniform over the disc" means probability is proportional to area. So the chance of landing within distance r of the centre is the area of the smaller disc divided by the area of the board, and that gives the distribution function of R straight away. Part (ii) then reduces to a single probability p per throw, a geometric-style "at least one success" calculation, and the minimisation of a function of an integer, which we do by looking at consecutive differences rather than by calculus.

Solution

Part (i)

Model each dart as a point uniformly distributed over a disc of radius a, so the probability that it lands in any region of the board is that region's area divided by πa^2.

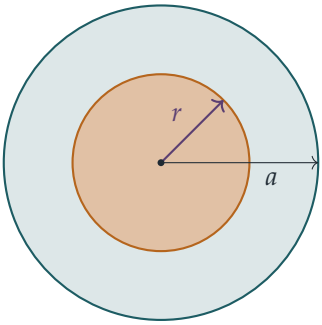
For $0 \leq r \leq a$, the event $R \leq r$ means the dart lands in the concentric disc of radius r, so

$$\mathrm{P}(R \leq r) = \frac{\pi r^2}{\pi a^2} = \frac{r^2}{a^2}.$$

Differentiating, R has probability density function

$$f(r) = \frac{2r}{a^2} \qquad (0 \leq r \leq a),$$

and $f(r) = 0$ otherwise.



Note that R is *not* uniform on $[0, a]$: there is more area (a longer thin ring) at larger radii, which is why the density grows like r.

Now

$$\mathrm{E}(R) = \int_0^a r \cdot \frac{2r}{a^2}\,\mathrm{d}r = \frac{2}{a^2} \cdot \frac{a^3}{3} = \frac{2a}{3},$$
$$\mathrm{E}(R^2) = \int_0^a r^2 \cdot \frac{2r}{a^2}\,\mathrm{d}r = \frac{2}{a^2} \cdot \frac{a^4}{4} = \frac{a^2}{2},$$

so

$$\mathrm{Var}(R) = \mathrm{E}(R^2) - (\mathrm{E}(R))^2 = \frac{a^2}{2} - \frac{4a^2}{9} = \frac{a^2}{18},$$

as required.

Part (ii)

(We read "with $a/\sqrt{10}$" as "within $a/\sqrt{10}$".) By part (i), the probability that a single throw lands within $a/\sqrt{10}$ of the centre is

$$p = \frac{(a/\sqrt{10})^2}{a^2} = \frac{1}{10}.$$

We assume the throws are independent (each is a fresh, uniformly random point on the board). Then the probability that none of the m throws is within $a/\sqrt{10}$ is $(0.9)^m$, so

$$\mathrm{P}(\text{he wins back £12}) = 1 - (0.9)^m.$$

His loss is the $£(12 + m)$ he pays minus anything he wins back, so his expected loss, in pounds, is

$$L(m) = 12 + m - 12(1 - (0.9)^m) = m + 12(0.9)^m.$$

Each extra throw costs £1 for certain but makes the £12 more likely; we need to find where these balance.

Ask yourself How do we minimise a function defined only on whole numbers?

Compare neighbouring values. For $m \geq 1$,

$$L(m+1) - L(m) = 1 + 12(0.9)^m(0.9 - 1) = 1 - 1.2\,(0.9)^m.$$

Since $(0.9)^m$ decreases as m increases, this difference is increasing in m, so L decreases while the difference is negative and increases once it is positive. The difference is negative exactly when $(0.9)^m > \frac{1}{1.2} = \frac{5}{6}$. Now

$$(0.9)^1 = 0.9 > \tfrac{5}{6}, \qquad (0.9)^2 = 0.81 < \tfrac{5}{6},$$

so $L(2) - L(1) < 0$, while $L(m+1) - L(m) > 0$ for every $m \geq 2$. Hence

$$L(1) > L(2) < L(3) < L(4) < \cdots,$$

and indeed $L(1) = 11.80$, $L(2) = 11.72$, $L(3) = 11.748$. (He is determined to play, so $m = 0$ is not allowed; in any case $L(0) = 12$ is larger.)

He should choose $m = 2$ throws, with an expected loss of £11.72. Since the minimum is strict, $m = 2$ is the only such value.

Checks

- A simulation of 4×10^5 uniform points in the unit disc gave mean 0.667, mean square 0.500 and variance 0.0556 $\approx \frac{1}{18}$, matching $\frac{2}{3}$, $\frac{1}{2}$ and $\frac{1}{18}$; about 10% of the points fell within $1/\sqrt{10}$ of the centre.
- Exactly, $L(1) = \frac{59}{5}$, $L(2) = \frac{293}{25}$, $L(3) = \frac{2937}{250}$, and a search over $1 \leq m \leq 99$ confirms $m = 2$ is the minimum.
- Sense check: the variance $a^2/18$ is far less than that of a uniform variable on $[0, a]$, namely $a^2/12$, because R is concentrated towards the rim.

Common slips

- Taking R to be uniform on $[0, a]$, which gives variance $a^2/12$.
- Forgetting the fixed £12 is only part of the cost: the loss is $12 + m$ minus the expected winnings, not just m.
- Setting the derivative of $m + 12(0.9)^m$ to zero gives $m \approx 2.3$, which is not an answer; you must compare the integer values on either side.

Beyond STEP

Finding the distribution of R from a uniform point in the plane is the simplest example of transforming a joint density into polar coordinates, where the Jacobian factor r in $\mathrm{d}A = r\,\mathrm{d}r\,\mathrm{d}\theta$ is exactly what makes the density of R proportional to r. Change of variables for joint densities, and optimisation over a discrete choice, are standard first-year probability topics. Grimmett and Welsh, *Probability: An Introduction*, covers both.

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